2 RELATED WORK

From Static Agents to Test-Time Learning. The majority of current AI agents, while capable, operate with static configurations that are manually designed and fixed after deployment (Wang et al., 2024; Xi et al., 2025; He et al., 2025b; Chen et al., 2025b;a; He et al., 2025c;d; Chen et al., 2025c). This limits their ability to adapt to novel situations, a key challenge motivating the development of "self-improving AI agents" (Gao et al., 2025b; Fang et al., 2025; Gao et al., 2025a; Sui et al., 2025;

2024b; Liu et al., 2025b). A prominent line of work enables agents to learn from past mistakes without updating weights. Reflexion (Shinn et al., 2023), a key baseline for our work, allows an agent to verbally reflect on trajectory failures and append these reflections to its prompt for subsequent episodes. Other approaches focus on enhancing agent memory. For instance, MemGPT (Packer et al., 2023) provides agents with a structured memory system to manage long contexts. Beyond reflection/memory, Uncertainty of Thoughts (Hu et al., 2024) adds test-time *uncertainty-aware planning*, deciding when to ask, verify, or revise without weight updates. While MemoryBank (Zhong et al., 2024) uses hierarchical summarization to retain information over long interactions.

Self-Evolving Agentic Systems. Another active area of research is the automated optimization of prompts that guide agent behavior (Liu et al., 2025a; Zhu et al., 2026; Hu et al., 2026). Generative approaches like APE (Zhou et al., 2022) and OPRO (Yang et al., 2023) use a powerful LLM to propose and score new prompts, iteratively refining them based on performance. Gradient-inspired methods like TextGrad (Yuksekgonul et al., 2024) refine prompts using LLM-generated textual feedback. Closely related to our work are evolutionary methods such as AlphaEvolve (Novikov et al., 2025), Promptbreeder (Fernando et al., 2023), and EvoPrompt (Guo et al., 2024), which maintain a population of prompts and apply genetic operators like mutation and crossover to discover more effective instructions. EvoTest generalizes prompt evolution to whole-system evolution, optimizing the entire agentic configuration—including the prompt, memory, hyperparameters, and tool-use routines. This allows for more holistic adaptations, such as tuning exploration strength, that are beyond the scope of prompt-editing alone. This vision for unified optimization is shared by EvoAgent (Yuan et al., 2024) and MASS (Zhou et al., 2025); Beyond 'Aha!' (Hu et al., 2025) complements this by aligning meta-abilities rather than only task prompts or single components.